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Title: Paediatric Arterial Line Insertion Policy

Introduction

Peripheral arterial access is an integral part of the management of many neonates and children in the Pediatric Intensive Care Unit. It allows medical staff to remove blood for investigations e.g. arterial blood gas or other haematology / biochemistry. Peripheral arterial access also allows for accurate invasive blood pressure monitoring thus providing information to tailor management e.g. the use of inotropes, and fluid boluses.

Aim

This guideline aims to outline the principles of management including insertion, sampling and removal of peripheral arterial access lines neonates and children admitted to the Pediatric Intensive Care unit, Pediatric Post Cardiac Surgery unit and Pediatric High Dependency units at Royal Hospital, Ministry of Health, Oman.

Abbreviations

ABG: Arterial Blood Gas

BP: Blood Pressure

PICU: Pediatric Intensive Care Unit

PPCSU: Pediatric Post Cardiac Surgery unit

PAL: Peripheral Arterial Line

Scope

Medical team in Pediatric critical areas including physicians and nurses.

Assessment

Indications:

- Hemodynamically unstable patients for beat to beat BP monitoring on inotropic support
- Patients undergoing major surgical procedures.

Contraindications:

1. Inadequate circulation to the extremity
2. Local skin infection
3. Malformation of limb
4. Recent cannulation or attempt of another artery in the same limb unless authorized by Senior Registrar/Consultant on service

Relative Contraindications:

1. Uncorrected Coagulopathy.

Complications:

1. Thrombo-embolism / vasospasm / thrombosis which can lead to compromise of arterial circulation, blanching, necrosis or gangrene of tissues or extremities
2. Damage to peripheral nerves
3. Emboli
4. Haematoma
5. Infection

If in doubt TAKE IT OUT

Sites for Peripheral Artery Catheters:

1. Radial artery
2. Posterior tibial artery
3. Dorsalis pedis artery
4. Ulnar artery (can be attempted after performing Allen's test if other sites are not obtainable)

The primary sites for peripheral arterial cannulation are the radial and posterior tibial arteries. The risk of ischemia secondary to radial or ulnar arterial cannulation is approximately 5%. The dorsalis pedis artery can be used if palpated (however it is absent in some infants). The brachial artery SHOULD NOT be used due to absence of collateral circulation.

Allen's Test: Only an artery with collateral circulation should be cannulated. Accessing collateral circulation can be achieved by performing the Allen's test if using an upper limb. The Allen's Test is a measurement of radial or ulnar patency. Performing the Allen's test in a neonate involves elevating the arm and simultaneously occluding the radial and ulnar arteries at the wrist, then rubbing the palm to cause blanching. Release the pressure on the ulnar artery. If normal colour returns to the palm in <10 seconds adequate ulnar circulation is present. Performing and reporting the results of the Allen's Test must be documented in the medical record.

Equipments:

1. Cleaned procedure trolley
2. Sterile dressing pack
3. Sterile gloves
4. Appropriate skin cleaning solution
5. 22-24G intravenous cannula (infants and neonates)
6. 20-22G intravenous cannula pediatric patients
7. 1-3ml syringe flushed with heparinised saline
8. Luer-lock, 10cm Connecter tube flushed with heparinised saline
9. Tapes &Tegaderm for securing the line
10. Splint / arm-board
11. Long extension line – minimum volume extension tubing
12. 50mls saline with Heparanised Saline (1unit/ml)
13. Transducer set and cable

Analgesia / Sedation:

This is a painful procedure. In most instances a neonate requiring an arterial line is unwell and will usually have fentanyl and/or midazolam infusions prescribed. A bolus may be required prior to commencement of the procedure.

In non- ventilated children consider local analgesia and procedural sedation when indicated.

Management:

Precautions:

- Avoid hyperextension of the joint as this may lead to occlusion of the artery
- Attempt insertion of no more than one artery. Seek senior help before further attempts
- Always ensure that the tips of the fingers / toes are exposed so that perfusion can be checked regularly

Intra-Arterial Fluids

Patency of the peripheral arterial line is maintained with an infusion of 0.9% sodium chloride for infusion with 1 unit of heparin per ml running at a rate of 0.5-1ml /hour. The Heparinised saline should be changed every 24 hours

NB – No other fluids / medications are to be given via an arterial line

Preparation and Procedure:

Before procedures:

- Wash hands and prepare work surface and equipment
- Perform Allen test to check for adequacy of collateral circulation (when indicated)
- Slightly extend the wrist / ankle to bring the artery closer to the surface
- Identify the artery by palpation +/- Doppler, USG.

During procedure:

- Rewash hands and put on gloves
- Clean the skin with Appropriate skin cleaning solution
- Insert the cannula over the artery at an angle of 30 - 45 degrees
- Puncture the artery and watch for blood in the hub of the cannula
- Withdraw the stylet whilst advancing the cannula slowly. There may be spasm from the artery having been touched hence blood return may be delayed
- Observe the cannula hub for pulsative blood flow
- Attach the cannula to the arterial connector and three-way tap and slowly flush with heparinised saline
- Turn off the three-way tap and secure the cannula with tape, and a splint
- Connect the10cm luer-lock, transducer set up and heparinised saline 50ml syringe

After procedure:

- Identify the line as an arterial line by placing a red arterial line sticker on the infusion line
- Observe the fingers / toes for circulation and warmth post procedure
- Level and zero the pressure transducer and set MEAN alarm limits as per medical preference
- Document blood pressure limits and target on treatment orders chart/ order sheet.
- Document procedure in the medical record
- Secure with splint and strapping as indicated.

Sampling from Peripheral Arterial Line:

Equipment needed:

- 2-5ml syringe

- Alcohol prep swab/ Chlorehexidine swabs when available
- Red IV cap

During procedure:

1. Wash hands and prepare work surface
2. Turn three-way tap 45 degrees ensuring no port will be open to air
3. Attach 2-5ml syringe to exit port of three-way tap
4. Turn three-way tap off to infusion line and open to patient and exit port then aspirate blood slowly from the patient into the syringe
5. Withdraw into the syringe enough blood to clear the arterial connector line of heparinised saline (2.0-5mls is usually sufficient)
6. Turn three way tap to 45 degrees ensuring no port will be open to air
7. Remove syringe from three-way tap and connect blood gas syringe / or syringe for blood sample collection
8. Turn three-way tap so that blood can be aspirated from neonate into the syringe
9. Once sufficient blood collection has occurred turn three-way tap 45 degrees ensuring no port will be open to air
10. Remove this syringe and replace with original syringe of aspirated blood and heparinised saline ensuring no air is present in the syringe
11. Slowly return the contents of this syringe back to the neonate observing for signs of arterial spasm such as blanching
12. Turn three way tap to 45 degrees ensuring no port will be open to air
13. Replace this syringe with 2ml syringe containing normal saline
14. Slowly flush the arterial connector and line with normal saline until the line is cleared of blood (2 -3mls is usually sufficient)
15. Turn three way tap to 45 degrees ensuring no port will be open to air

After procedure:

1. Clean the exit port with alcohol swab after removing the syringe and clean the port opening
2. Once cleaned close the port opening with the new red IV cap
3. Turn three-way tap open to infusion of heparinised saline and patient and closed to the exit port and commence infusion

4.Observe digits and limb for colour, warmth and circulation

5.Send blood samples as requested

Daily care of an Arterial line:

- The need for an arterial line should be reviewed and addressed daily, if it is no longer needed it should be removed.
- The physician as well as the nurse should inspect the site of the line and look for any signs of infection or ischemia.

What to do in case of signs of Ischemia:

- The line should be removed immediately.
- After consulting with the senior physician an infusion of heparin may be started 10 IU/kg/hr and to monitor the signs of ischemia and closely monitor the pulses.
- Consider doing USG Doppler of the affected limb to document the presence of thrombosis. It will also help as a tool for follow up.
- Consider also applying Glycerl trinitrite patch or cream to the affected limb.

Nursing Management:

- Careful positioning of the limb with the PAL in-situ
- Ensure the tips of the fingers / toes are exposed at all times
- Observe for adequate patency of artery by ensuring digits and limb are pink, warm and well perfused
- Inspect and document hourly the area distal and proximal to the insertion site for blanching, redness, cyanosis and changes in temperature and perfusion
- Report to In-charge nurse/ medical team if there are changes to circulation of the limb or digits
- Level and zero arterial line at commencement of every shift and if patient is turned / moved
- Heparinised saline only to be administered as a continual infusion at between 0.5 – 1.0ml/hour
- Change heparinised saline infusion syringe every 24 hours and infusion line every third day
- Arterial line is only to be used for blood sampling and blood pressure monitoring

1. Wash hands and prepare work surface
2. Switch off heparinised saline infusion
3. Remove tapes carefully to ensure skin integrity
4. Withdraw the cannula and apply pressure to the site for 5 minutes with a piece of sterile gauze / cotton ball ensuring circulation to the hand / foot is maintained
5. Check to see if the bleeding has stopped. If it has not apply pressure for a further 2 – 3 minutes before checking again – repeat this step as needed
6. Once bleeding has stopped cover the site with a small piece of gauze and tape / film dressing
7. Observe the digits and limb for adequate circulation and continue to monitor for the next four hours
8. Document procedure in the medical record.

Disclaimer:

Guidelines are general and cannot take into account all of the circumstances of a particular patient. Judgment regarding the propriety of using an specific procedure or guideline with a particular patient remains with the patient's physician, nurse, or other health care professional, taking into account the individual circumstances presented by the patient.

References:

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Written By: Dr.Balqees Al Noobi
Checked By: PICU Team
Authorized by: KHOULA JULENDA AL SAID

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